

“Where Does the Model Look?” — Attention Across Modalities

Neya Balakrishna, Aniket Iyer, Avani Hanasoge, Raj Patel

Department of Electrical & Computer Engineering

Georgia Institute of Technology

{nbalakrishna3, aiyer392, ahanasoge6, rpatel627}@gatech.edu

Abstract

Modern vision-language models (VLMs) increasingly support multimodal dialogue, yet their ability to integrate visual evidence with evolving conversational context remains poorly understood. This project introduces a unified framework for analyzing how three leading architectures—BLIP-2, LLaVA-1.5, and Qwen2.5-VL—reason over images in both single-turn and multi-turn settings. Using COCO images, we construct controlled prompts containing accurate captions, deliberately altered captions to probe caption sensitivity, and a Visual Necessity Ladder of questions. Each model answers these prompts in both conversational modes, allowing us to assess how dialogue history shapes multimodal reasoning. We evaluate models with attention-based metrics and an external multimodal judge to measure visual grounding, susceptibility to linguistic bias, and error propagation across turns. Our results reveal when conversational history strengthens or weakens visual grounding, when linguistic priming overrides perception, and how these effects differ across architectures. (Rohrbach et al., 2019) The framework offers a principled method for evaluating VLMs as conversational multimodal agents rather than static image-question systems.

1 Introduction

1.1 General Context and Background

Vision-language models (VLMs) have rapidly expanded the capabilities of modern artificial intelligence by enabling systems to jointly process images and text. These models now support tasks

such as captioning, visual question answering, and multimodal reasoning, which require coordinated use of linguistic and perceptual information. Recent VLMs also operate within interactive dialogue settings, allowing users to query images through natural conversation. This shift from static, single-turn queries to dynamic, multi-turn dialogue has increased interest in how VLMs track context, integrate visual evidence, and maintain grounding over time.

Despite these advances, the internal mechanisms that guide cross-modal integration remain only partially understood. In particular, attention flows inside decoder-based VLMs can reveal how strongly a model relies on visual features compared to linguistic cues, how its reasoning changes with question difficulty, and how conversational history affects its decision-making. Understanding these mechanisms is essential for building trustworthy multimodal dialogue systems, especially as VLMs become embedded in user-facing applications.

To address these gaps, this project will systematically analyze how attention propagates and redistributes across image and text in three representative architectures: BLIP-2, LLaVA-1.5 and Qwen. By applying attention rollout (Abnar and Zuidema, 2020), we trace information flow through successive transformer layers and quantify fusion behavior using standardized measures—Modality Dominance Index, attention entropy, and layer-wise attention shift. (Chefer et al., 2021) Evaluations on matched and mismatched image-caption pairs will probe whether greater fusion strength leads to more semantically balanced and interpretable attention, shedding new light on the internal dynamics and failure cases of multimodal AI systems.

1.2 The Problem and the Gap

Most existing evaluations of VLMs focus on single-turn question–image pairs. These settings ignore the conversational structure through which real users interact with multimodal systems. As a result, several open questions remain. It is unclear how much VLMs rely on textual priors when earlier conversational turns contain misleading or incorrect information. It is also unclear whether visual grounding improves or degrades as the dialogue progresses. Furthermore, current benchmarks rarely compare different architectures under controlled conditions that separate the influence of captions from genuine visual recognition.

This gap is significant because conversational agents can accumulate errors across turns. A misleading caption or an incorrect earlier response may bias future reasoning, which can amplify hallucination, reduce grounding, or distort attention allocation. Without controlled evaluation of these effects, it is difficult to assess the reliability of VLMs in realistic interactive environments.

1.3 Aims and Objectives

The purpose of this study is to develop a controlled and reproducible framework for analyzing multimodal reasoning in both single-turn and multi-turn conversational settings. The study has four main objectives.

- To measure how three leading VLMs, BLIP-2, LLaVA-1.5, and Qwen2.5-VL, respond to accurate and inaccurate captions.
- To evaluate how visual grounding changes as question difficulty increases along a Visual Necessity Ladder.
- To quantify the effect of conversational history on hallucination, reliance on textual priors, and cross-modal attention.
- To compare architectural differences by examining attention patterns and correctness across conditions.

1.4 Significance and Contribution

This work introduces a unified methodology that combines controlled image-based stimuli, systematic caption manipulation, and a difficulty-graded question ladder with attention-centered analysis. The approach allows direct comparison between single-turn and multi-turn reasoning, exposing

how conversational context influences multimodal grounding. The study contributes new insights into how different VLM architectures balance visual and linguistic signals, how misleading information propagates through dialogue, and when attention shifts signal grounding failures. The findings support both theoretical understanding of cross-modal attention and practical guidelines for designing robust conversational AI systems.

The remainder of the paper is structured as follows. Section 2 reviews related work. Section 3 describes the dataset construction, experimental conditions, and analysis pipeline. Section 4 presents the results, followed by discussion in Section 5. Section 6 concludes with implications and directions for future research.

2 Previous Work

Early multimodal systems often relied on tightly coupled vision and language encoders, which limited scalability and transferability. Recent work has focused on modular architectures that combine pretrained vision backbones with large language models through learned bridging components. BLIP-2 is a representative example of this approach. It uses a frozen image encoder and a frozen language model connected by a lightweight Q-former that translates visual features into a sequence of tokens that the language model can process. This design greatly reduces training cost and supports flexible integration with different LLMs. A known limitation is that BLIP-2 tends to over rely on textual priors when visual evidence is ambiguous. This over reliance often manifests as hallucination or object insertion. These limitations motivate exploring architectures that incorporate stronger vision alignment and instruction tuning. (Li et al., 2023)

LLaVA-1.5 extends the BLIP-2 paradigm by incorporating multimodal instruction tuning. It uses a CLIP based visual encoder and aligns visual features directly to a language model using a projection layer. The key innovation is large scale conversational finetuning that teaches the system to follow multimodal instructions. This results in stronger interactive reasoning and more stable question answering in dialogue settings. However, LLaVA remains sensitive to misleading linguistic context and sometimes struggles with detailed visual grounding. These issues encourage the examination of models that embed visual tokens more

tightly within the language model’s token space. (Liu et al., 2024)

Qwen2.5-VL represents a newer class of models where visual information is discretized into visual tokens and processed through the language model in a unified token space. This token unification improves cross modal attention, supports fine grained grounding, and enables more coherent multi turn dialogue behavior. Although promising, Qwen still exhibits susceptibility to linguistic drift over long contexts. These observations collectively motivate the need for systematic evaluation of conversational visual grounding across multiple architectures. (Bai et al., 2025)

3 Experimental Setup

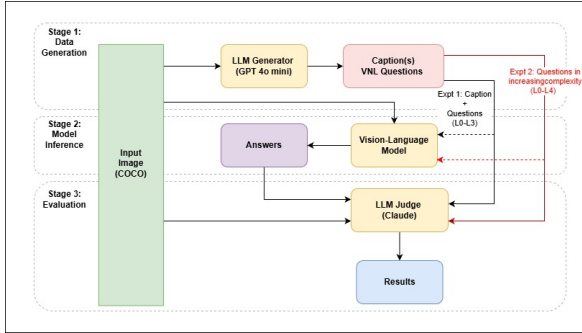


Figure 1: Experimental pipeline for generating questions, producing model answers, and evaluating outputs across caption conditions and difficulty levels.

3.1 Experiment 1: Caption Sensitivity & Hallucination Susceptibility

3.1.1 Objective

Experiment 1 evaluates how vision–language models (VLMs) respond to accurate versus misleading textual context and how strongly captions influence visual reasoning. The core question is: *When a model receives an incorrect description of an image, does it recognize the inconsistency using visual evidence, or does it accept the textual cue as truth?*

The experiment also compares this behavior in single-turn and multi-turn settings. In single-turn mode, each question is paired only with its caption, allowing caption effects to be isolated. In multi-turn mode, the caption becomes the first conversational turn, and subsequent questions are added sequentially. This setup tests whether misleading information becomes more influential

when reinforced by the ongoing dialogue and whether models accumulate errors across turns.

A final objective is to examine how architectural differences influence caption sensitivity. Because BLIP-2, LLaVA-1.5, and Qwen2.5-VL use distinct cross-modal alignment mechanisms, they may differ in their reliance on textual cues versus visual evidence. Evaluating all three under identical conditions allows us to identify which architectural choices strengthen or weaken grounding under inaccurate captions.

Together, these objectives make Experiment 1 a systematic test of how VLMs balance visual evidence against textual priors, how conversational history shapes this balance, and how architecture affects robustness to caption-driven hallucination.

3.1.2 Pipeline

Experiment 1 uses a controlled caption-manipulation pipeline built on COCO images. For each image, the system generates two captions: an accurate caption describing the scene correctly, and a manipulated caption that is identical except for two systematically altered visual attributes (e.g., object type, color, action, or spatial relation). This ensures that the incorrect caption conflicts with the image in a clear and measurable way.

Next, three evaluation questions are produced using a constrained template: L0 is a language-only question that does not require the image. L1 and L2 are visual questions targeting the two altered attributes. L3 is a higher-level visual reasoning question that references the image but does not depend on the manipulated attributes. This progression allows us to test caption effects across increasing levels of visual necessity.

The experiment is conducted in single-turn and multi-turn modes. In single-turn mode, each caption–question pair is evaluated independently. In multi-turn mode, the caption forms the first dialogue turn, and questions are introduced sequentially, preserving the model’s previous answers as part of the conversational context. All three models—BLIP-2, LLaVA-1.5, and Qwen2.5-VL—receive the same images, captions, and questions.

3.1.3 Methodology

During inference, the model produces an answer and decoder attention maps. These maps are collected across layers and used to compute three attention-centered metrics:

- **Modality Dominance Index (MDI)** – measures whether attention concentrates more on visual or textual tokens.
- **Attention Entropy** – reflects how sharply attention is distributed.
- **Attention Shift** – captures how the model’s focus changes from one turn to the next.

In addition to attention-based measures, we compute two outcome-based metrics: **overall accuracy** and **fooling rate**. Fooling rate is defined as the proportion of cases in which the model answers correctly with the accurate caption but answers incorrectly when given the misleading caption. This metric provides the clearest indicator of caption-driven hallucination.

Together, these metrics quantify both the model’s internal grounding dynamics (via attention) and its external behavioral robustness under misleading language.

To evaluate correctness, an external multimodal judge receives the image, question, and model response. The judge assigns labels without access to the caption, ensuring that the evaluation reflects true visual understanding. These labels allow measurement of accuracy, hallucination rates, and susceptibility to misleading captions.

By comparing results across models, dialogue modes, and difficulty levels, the experiment provides a systematic view of how language bias, architecture, and conversational structure interact to shape multimodal reasoning.

3.2 Experiment 2: Visual Necessity and Reasoning Complexity

3.2.1 Objective

Experiment 2 examines how models transition from generic world knowledge to image-grounded reasoning as questions become increasingly dependent on visual evidence. Each image is paired with a five-level question ladder (L0–L4). L0 asks a general question whose answer does not require the image, while L1–L4 require progressively more specific visual attributes such as object type, color, action, spatial relation, or interaction between entities. This structure allows us to test whether models move beyond generic answers at L0 and incorporate actual visual details when later questions demand them. A second objective is to study how reasoning behavior evolves across turns in single-turn and multi-turn settings. In the

multi-turn mode, early answers remain in the conversational context, which may reinforce priors or propagate early mistakes. A final objective is to determine how architectural differences influence this transition, given that BLIP-2, LLaVA-1.5, and Qwen2.5-VL integrate visual features in different ways.

3.2.2 Pipeline

Experiment 2 uses the same COCO images as Experiment 1 but manipulates the questions instead of the captions. For each image, an accurate caption is passed to a question-generation module that produces a structured five-step ladder (L0–L4). L0 is a general world-knowledge question with a textbook-style answer; L1–L4 are image-dependent questions whose required specificity increases at each level. As in Experiment 1, the experiment is conducted in both single-turn and multi-turn modes. In the single-turn condition, each ladder question is asked independently. In the multi-turn condition, the ladder is introduced sequentially as a short dialogue, starting with L0 and ending with L4, with the model’s previous answers preserved in the conversational history. This design allows us to observe whether conversational context stabilizes image-grounded reasoning or amplifies generic priors and early errors. All models receive the same images, captions, and question ladders, and all answers are evaluated using the same external multimodal judge as in Experiment 1.

3.2.3 Methodology

For each image and ladder level, the model produces an answer and decoder attention maps. Attention maps are collected across layers and used to compute Modality Dominance Index, Attention Entropy, and Attention Shift. These metrics capture how attention allocation shifts from reliance on textual priors to reliance on visual features as the question demands more detailed image understanding. We also measure accuracy at each ladder level (L0–L4) to quantify how consistently the model transitions from general knowledge to image-specific reasoning. This combination of attention-based and outcome-based measures provides insight into how deeply the model grounds its answers in the visual input as reasoning complexity increases.

3.2.4 Hypotheses

We hypothesize that models will initially rely heavily on generic world knowledge for L0 questions, reflecting low visual grounding. As questions become more visually specific at L1–L4, models with stronger grounding mechanisms should increasingly incorporate visual evidence, while models that depend more on priors may continue producing generic answers. We also expect multi-turn prompting to amplify early errors or assumptions, allowing incorrect L0 answers to persist through later turns and reduce image-specific accuracy. Finally, we hypothesize that architectural differences will influence these effects. BLIP-2 may adapt more slowly to image details due to its Q-former bottleneck, while LLaVA and Qwen, which align visual features more directly with the language model, may transition more effectively from generic to image-grounded reasoning as visual necessity increases.

4 Evaluations

4.1 Overview of Metrics

To test whether attention reflects *true semantic grounding*, each model will be evaluated on both matched (true) and mismatched (negative) image–caption pairs. Mismatched pairs are generated by swapping captions across unrelated images.

Grounded models should exhibit coherent, localized attention for matched pairs and diffuse or inconsistent focus for mismatched ones. This difference will be captured through three quantitative metrics:

1. **Modality Dominance Index (MDI)** — measures relative attention toward visual vs. textual tokens:

$$\text{MDI} = \frac{\sum_{i \in \text{Vis}} \sum_j \tilde{A}_{ij}}{\sum_{i,j} \tilde{A}_{ij}}, \quad (1)$$

where Vis indexes visual tokens. Higher MDI for matched pairs indicates stronger visual grounding; lower values in mismatched cases imply text-dominant bias.

2. **Attention Entropy (H)** — quantifies attention sharpness per token:

$$H = -\frac{1}{N} \sum_i \sum_j \tilde{A}_{ij} \log \tilde{A}_{ij}. \quad (2)$$

Lower entropy corresponds to confident, localized attention (matched pairs); higher entropy reflects diffuse focus (mismatched).

3. **Layer-wise Attention Shift (ΔA)** — measures how attention evolves through depth:

$$\Delta A^{(l)} = \frac{\|\tilde{A}^{(l)} - \tilde{A}^{(l-1)}\|_1}{N^2} \quad (3)$$

Larger shifts for matched inputs suggest active reallocation toward semantically relevant regions.

4. **Fooling Rate** — defined as the fraction of cases where a model answers correctly under the accurate caption but becomes incorrect when the caption is corrupted

$$\text{FoolRate} = \frac{1}{N} \sum_{i=1}^N I_i, \quad (4)$$

with

$$I_i = \begin{cases} 1, & \text{if } g_i(\hat{y}_i^{\text{corr}}) = 1 \text{ and } g_i(\hat{y}_i^{\text{incorr}}) = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

where

N number of evaluated image–question pairs;

$\hat{y}_i^{\text{corr}}$ model answer for example i with the correct caption;

$\hat{y}_i^{\text{incorr}}$ model answer for example i with the corrupted caption;

$g_i(\hat{y}) \in \{0, 1\}$ external judge for example i (1 = correct, 0 = incorrect);

$I_i \in \{0, 1\}$ indicator (1 = model is correct with the correct caption and wrong with the incorrect caption; 0 otherwise).

Modify this: adding fooling rate to this para:

Together, these metrics quantify bias (MDI), selectivity (H), and adaptivity (ΔA), providing a unified framework for comparing attention structure and grounding strength across architectures.

4.2 Results of Experiment 1

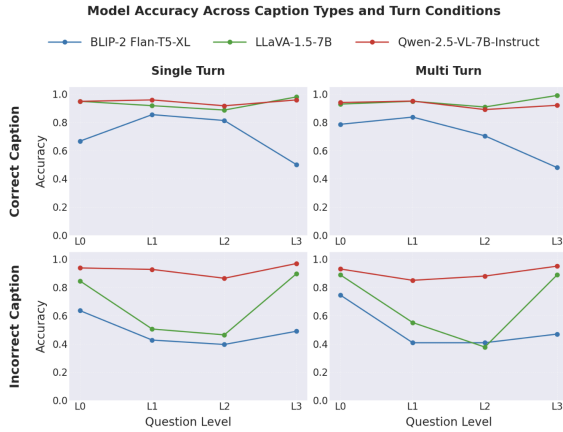
Caption Sensitivity & Hallucination Susceptibility

4.2.1 Accuracy

With correct captions, accuracy remains high in both single-turn and multi-turn settings; multi-turn dialogue is comparable or slightly better, indicating that accurate textual context does not harm performance.

With incorrect captions, accuracy drops substantially, especially on visually grounded questions (L1/L2). Under incorrect captions, multi-turn dialogue often further reduces accuracy, suggesting that repeated exposure to a misleading description reinforces caption-driven errors across turns.

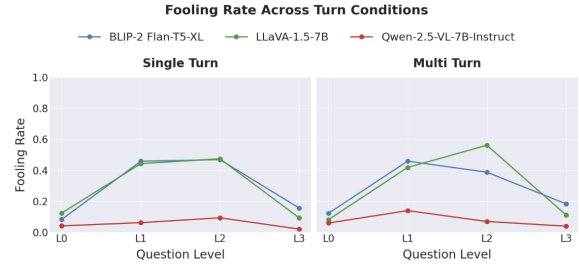
Qwen2.5-VL is consistently the most accurate and robust to incorrect captions, LLaVA-1.5 is intermediate and degrades under misleading text, and BLIP-2 is the weakest, especially with incorrect captions and in multi-turn dialogue. This aligns with their architectures: Qwen2.5-VL processes images as visual tokens inside a unified language-token space (supporting tighter grounding), whereas BLIP-2 relies on frozen vision and language backbones connected through a Q-former bottleneck, which in our experiments corresponds to lower accuracy under misleading captions.



4.2.2 Fooling Rate

Across all models, fooling rates are low on the language-only L0 questions and peak on the mid-ladder, visually grounded questions L1–L2, showing that wrong captions are most damaging when they change concrete visual details (e.g., object, color, action). Multi-turn dialogue generally increases fooling rates relative to single-turn, indicating that repeated exposure to an incorrect caption reinforces caption-driven errors over time.

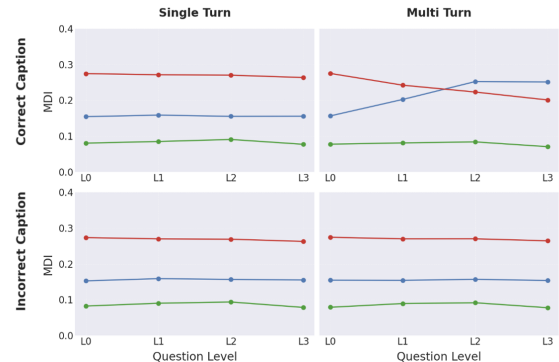
Qwen2.5-VL remains the most robust, with fooling rates staying low in both single- and multi-turn modes. BLIP-2 and LLaVA-1.5 are both substantially more vulnerable to misleading captions, with fooling rates peaking on L1–L2 questions, which ask about the manipulated visual attributes (e.g., object type or color). LLaVA-1.5 is slightly more affected in some multi-turn cases, which is consistent with its strong language priors from multimodal instruction tuning.



4.2.3 Modality Dominance Index (MDI)

Across all conditions, Qwen2.5-VL has the highest MDI, BLIP-2 is in the middle, and LLaVA-1.5 has the lowest, indicating that Qwen attends most to image tokens and LLaVA relies most on text. In single-turn mode, MDI is fairly flat across L0–L3 for all three models and barely changes between correct vs. incorrect captions, suggesting that caption corruption mainly affects answers rather than how much attention is put on the image. In multi-turn mode with correct captions, BLIP-2 shows a clear rise in MDI as questions become more visual, while Qwen starts with high visual focus and decreases slightly; LLaVA stays low and almost unchanged. With incorrect captions, these trends become weaker, but Qwen still allocates the most visual attention, whereas LLaVA remains strongly language-dominated.

Modality Dominance Index (MDI) Across Caption Types and Turn Types



4.3 Results of Experiment 2

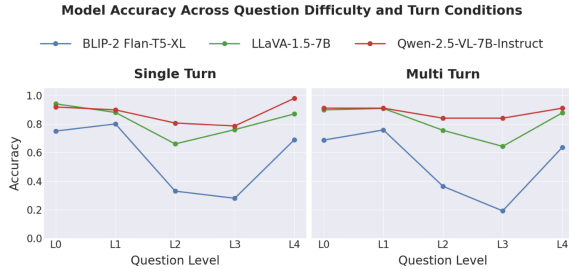
Visual Necessity and Reasoning Complexity

4.3.1 Accuracy

For both single-turn and multi-turn prompting, L0 is consistently easiest, and accuracy decreases on mid-ladder, image-dependent questions (L2/L3) before partially recovering at L4.

Multi-turn accuracy is generally close to single-turn accuracy: it sometimes provides a modest benefit on easier levels, but on the most visually demanding levels it can lead to additional degradation, indicating that early priors may influence later answers.

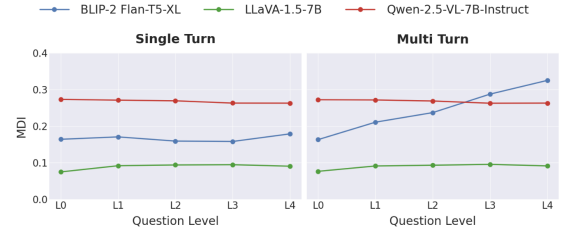
Qwen2.5-VL is strongest, likely helped by its unified visual-token space inside the language model, which supports stable image grounding across levels. LLaVA-1.5 performs moderately well but drops on harder visual questions, consistent with its reliance on instruction-tuned text priors. BLIP-2 is weakest, especially in multi-turn and mid-ladder levels, which matches its more fragile grounding from a frozen vision encoder passed through a narrow Q-former bottleneck.



4.3.2 Modality Dominance Index (MDI)

Single-turn MDI stays fairly flat: Qwen2.5-VL pays the most attention to image tokens, BLIP-2 is in the middle, and LLaVA-1.5 relies most on text. In the multi-turn setting, BLIP-2's MDI increases with question level, meaning it looks at the image more as questions become harder. Qwen2.5-VL is strongly image-grounded in both modes, while LLaVA-1.5 remains more text-driven even for the more visual questions.

Modality Dominance Index (MDI) Across Question Difficulty and Turn Conditions



4.4 Exploratory Metrics: Attention Shift & Entropy (Inconclusive)

In addition to accuracy and Modality Dominance Index, we initially planned to analyze two exploratory attention based metrics: turn level attention shift and attention entropy. For all three models, we collected decoder attention tensors and implemented helper functions that flattened layer wise attention, averaged over batch and heads, and extracted the final token's attention distribution as a proxy for the model's focus over the joint visual textual context. For LLaVA in particular, we implemented a normalized entropy measure over this final distribution, with the goal of tracking how sharply attention concentrated as questions became more visually demanding. Similar utilities were sketched for computing per turn attention shift by comparing attention rows across sequential questions.

In practice, these metrics proved numerically and conceptually unstable. The entropy computation occasionally produced invalid values, including NaNs, due to edge cases where the summed attention mass was extremely small or zero after averaging and masking, which led to division by very small values and log operations near zero. Differences in attention tensor formats across architectures further complicated matters. LLaVA returns nested tuples of attention tensors, BLIP and Qwen expose different attention shapes, and some layers focus almost entirely on special or newly generated tokens, which causes entropy values to collapse to trivial extremes that are difficult to interpret. For attention shift, the growing sequence length across turns in the multi turn setting makes it non trivial to align attention vectors, since later turns contain additional tokens that have no direct counterpart in earlier distributions. Without careful alignment to a common prefix, the apparent shift mostly reflects the presence of new tokens rather than a meaningful reallocation of attention over the same context.

4.5 Cross-Experiment Summary

Accuracy

Caption errors in Experiment 1 have a stronger effect on accuracy than increased visual difficulty in Experiment 2; switching from a correct to an incorrect caption produces a larger accuracy drop than moving from L0 to higher ladder levels.

Multi-turn interaction magnifies caption-induced errors more clearly in Experiment 1 than it magnifies difficulty effects in Experiment 2, highlighting the particular risk of misleading textual context.

Across both experiments, models perform best on generic L0 questions and are least reliable on visually demanding questions, indicating persistent difficulty in shifting from language-based priors to robust image-grounded reasoning.

TO DO: Avani: Add summary for MDI across exp 1 and exp 2

5 Conclusion and Future Direction

TO DO: Avani

References

- Samira Abnar and Willem Zuidema. 2020. <https://doi.org/10.18653/v1/2020.acl-main.385> Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4190–4197, Online. Association for Computational Linguistics.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibor Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025. <https://arxiv.org/abs/2502.13923> Qwen2.5-vl technical report. *Preprint*, arXiv:2502.13923.
- Hila Chefer, Shir Gur, and Lior Wolf. 2021. <https://arxiv.org/abs/2103.15679> Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. *Preprint*, arXiv:2103.15679.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. <https://arxiv.org/abs/2301.12597> Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *Preprint*, arXiv:2301.12597.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. 2024. <https://arxiv.org/abs/2310.03744> Improved baselines with visual instruction tuning. *Preprint*, arXiv:2310.03744.

Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. 2019. <https://arxiv.org/abs/1809.02156> Object hallucination in image captioning. *Preprint*, arXiv:1809.02156.